

From Clicks to Carts: Reinforcement Learning for Intent-Driven Personalized Product Search in E-Commerce

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Abstract

In E-commerce search systems, relevance and engagement signals alone may not fully capture differences in purchase intent across user interactions. We propose Intent-Driven LLM Query Reformulator (IDLQR), a framework that uses reinforcement learning to optimize the customer journey such that products added to cart are ranked above those merely clicked.

IDLQR employs a two-stage approach: first, a large LLM teacher (DeepSeek-R1-Distill-Qwen-32B) generates high-quality query rewrites by reasoning over serialized user sessions including clicks, add-to-cart actions, and product metadata that are grounded in shopping intent. These rewrites are distilled into a compact student model (Qwen-0.6B) for real-time serving. The student is then fine-tuned with Group Relative Policy Optimization (GRPO) using a novel reward function that explicitly prioritizes conversion signals over engagement signals. Experiments on a large-scale proprietary dataset (95K customers, 820K sessions) show that IDLQR achieves parity with zero-shot models up to 13× larger while requiring a fraction of the computational resources, achieving a 7.9% MAP@1 and 8.3% MAP@3 improvement while maintaining sub-100ms latency suitable for production deployment. These offline results suggest that task-specific distillation combined with reward-driven optimization can provide a competitive alternative to model scaling for latency-constrained e-commerce search.

Keywords

e-commerce search, recommendation systems, intent-aware ranking, LLMs, query rewriting, reinforcement learning, GRPO, knowledge distillation

1 Introduction

The modern e-commerce landscape presents unprecedented challenges in understanding and optimizing customer journeys. E-commerce search systems commonly use relevance and engagement signals to optimize retrieval and ranking [12]. However, these signals may not fully reflect differences in purchase intent across customer interactions. Query rewriting addresses the lexical gap between user queries and product descriptions [7, 13]. Existing approaches include discriminative models with limited flexibility, and

generative LLM-based methods that produce high-quality rewrites [3] but face significant inference latency for online deployment. These limitations often force offline deployment, making systems vulnerable to semantic drift in dynamic e-commerce environments [5, 18].

Recent advances in reinforcement learning for search systems have shown promising results in optimizing long-term engagement [9, 16], while knowledge distillation has proven effective in creating efficient student models from larger teachers [8]. Despite these advances, conversion-oriented objectives for personalized query reformulation remain comparatively underexplored. Relevance and engagement signals alone may not fully capture differences in purchase intent across user interactions. In e-commerce, users may click multiple products before progressing toward conversion, motivating objectives that more explicitly account for the strength of behavioral signals along the customer journey.

This paper proposes a novel framework combining LLM-based query rewriting with knowledge distillation and reinforcement learning to optimize for conversion. Our approach recognizes that cart additions represent stronger purchase intent than clicks, training models to promote conversion-oriented results. Our contributions are methodological and empirical: we introduce a conversion-oriented reward formulation for GRPO-based query reformulation, develop a compact teacher-student architecture suitable for latency-constrained serving, and evaluate the resulting model on large-scale real-world e-commerce interaction data.

2 Related Work

Query Reformulation in E-Commerce. Classical information retrieval relies on lexical matching methods like BM25 [4], with query expansion techniques addressing vocabulary mismatch [7, 13]. However, these methods fail to capture evolving user intent across multi-step sessions [11, 23]. The emergence of LLMs has advanced query reformulation where the *Rewrite-Retrieve-Read* paradigm demonstrates that LLM-generated rewrites can outperform human reformulations [15]. However, these approaches primarily rely on supervised fine-tuning or static prompting, with limited exploration of reinforcement learning for optimizing conversion.

LLMs for Recommendation and Search. LLMs are reshaping recommendation systems by shifting from ID-based embeddings to text-centric representations [6, 19, 24]. Another research strand

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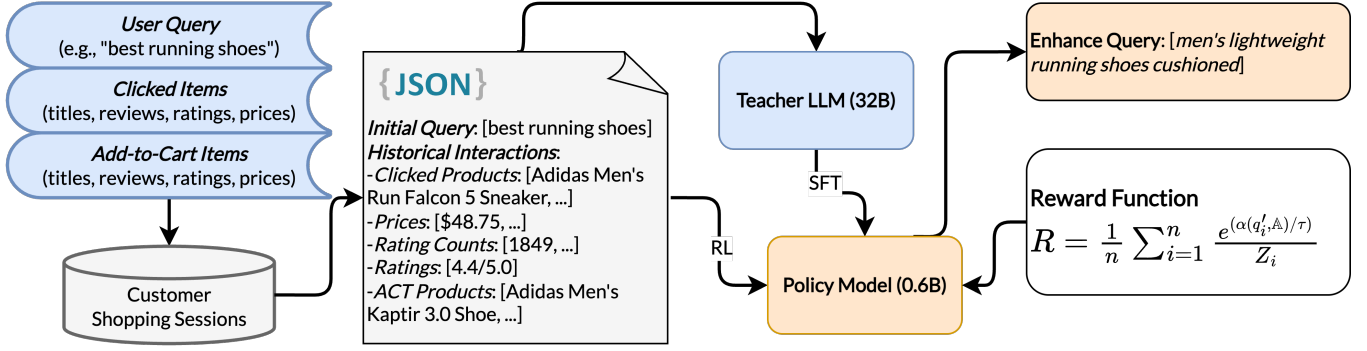


Figure 1: System architecture showing the two-stage training process: knowledge distillation from teacher to student, followed by reinforcement learning optimization using GRPO with conversion-focused rewards.

models user behavior sequences as natural language such as CXSimulator [14] that demonstrates how event sequences can be serialized into textual narratives, enabling plausible user behavior simulations. Incorporating behavioral signals as textual prompts has shown promise in improving personalization and retrieval quality [17, 25].

Reinforcement Learning for Search Optimization. Reinforcement learning aligns LLM outputs with downstream objectives, most prominently through Proximal Policy Optimization (PPO) [20]. However, PPO’s reliance on critic networks introduces computational overhead. Group Relative Policy Optimization (GRPO) [21, 27] addresses these challenges by normalizing candidate rewards against the group mean, eliminating separate value networks. This critic-free approach provides memory efficiency suitable for large-scale LLM training, while its capacity to incorporate business-oriented rewards makes it promising for conversion-aware query reformulation [1].

Research Gaps. Despite these advances, the application of reinforcement learning to conversion-oriented e-commerce query reformulation remains comparatively underexplored [1]. This motivates a framework that combines efficient query reformulation with an objective designed to account for behavioral signals associated with different stages of the customer journey.

3 Methodology

Our approach, Intent-Driven LLM Query Reformulator (IDLQR), employs a two-stage training framework combining knowledge distillation from large teacher models with GRPO-based reinforcement learning. The core innovation lies in explicitly designing rewards that prioritize conversion signals (add-to-cart) over engagement signals (clicks).

3.1 Architecture Overview

The system (Figure 1) comprises five components: (1) a **Teacher Model** (DeepSeek-R1-Distill-Qwen-32B) generating high-quality rewrites from ground-truth conversion data; (2) a **Student SFT** (Qwen-0.6B) trained via supervised fine-tuning to mimic teacher behavior; (3) a **Student RL** stage further optimizing the student with GRPO; (4) a **Relevance Scorer** for reward computation; and (5) a **Deployment** module serving real-time rewrites in production.

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## Context:
You are an expert e-commerce search optimization assistant. Your task is to
analyze customer behavior patterns and enhance their search query to better
align with their emerging preferences and shopping intent.

## Input Data:
Original Query: [{row.get("search_query", "")}]
Product Interactions:
- Clicked Products: [{row.get("clicked_item_names", "")}]
- Prices: [{row.get("clicked_prices", "")}]
- Rating Counts: [{row.get("clicked_cr_total_rating_count", "")}]
- Ratings: [{row.get("clicked_cr_display_rating", "")}]

## Query Enhancement Guidelines:
1. Analyze attribute frequency:
- Only include attributes present in >80% of viewed/carted items
- Minimum of 3 product interactions required before considering a pattern
- Discount outlier interactions that deviate significantly from the main pattern
2. Prioritize dominant patterns:
- Price range (if >80% of interactions fall within a specific range)
- Style elements (if consistently present across >80% of viewed items)
- Features/specifications (if selected in >80% of interactions)
- Brand preferences (if >80% of interactions involve specific brands)
3. Maintain search breadth:
- Add no more than 2-3 high-confidence attributes to the original query
- Preserve the core search intent from the original query
- Avoid over-specification that might exclude relevant results

## Output Format:
Original Query: [{row.get("search_query", "")}]
Enhanced Query: [Optimized query with validated pattern attributes]
Confidence: [High/Medium/Low based on pattern strength]
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Figure 2: Prompt template instructing the LLM to enhance the customer’s search query to capture evolving shopping intent while avoiding over-specificity.

3.2 Query Reformulation with LLMs

Unlike embedding-based methods that compress interactions into dense vectors [2, 10], our method uses LLM reasoning over structured textual narratives that preserve temporal dynamics [26]. A user session $S = \{(a_t, p_t, \tau_t)\}_{t=1}^T$ is serialized into a natural language prompt: each interaction type a_t (search, click, add-to-cart), selectively summarized product metadata p_t (title, brand, key attributes), and temporal signal τ_t are translated into human-readable sentences. For long sessions, recency-weighted truncation prioritizes the most recent interactions. The complete input $x = [\text{serialize}(S), q_0]$ concatenates the session narrative with

InputOriginal Query (q): *black dress***Clicked Items ($\mathbb{C}$):**

- Casual t-shirt dress (★4.2, 5,200 reviews, \$22)
- Sleeveless midi sundress (★4.4, 3,100 reviews, \$29)
- Smocked cocktail dress (★4.3, 2,700 reviews, \$34)
- Square-neck midi dress (★4.3, 2,100 reviews, \$31)
- Ruffle wrap mini dress (★4.2, 1,900 reviews, \$27)

OutputReformulated Query (q'):
*black midi cocktail dress elegant party wear under \$35***Groundtruth****Add-to-Cart Items ($\mathbb{A}$):**

- Smocked cocktail dress (★4.3, 2,700 reviews, \$34)
- Sleeveless midi sundress (★4.4, 3,100 reviews, \$29)

Figure 3: The RL process shifts a generic *black dress* query toward one reflecting purchase intent by analyzing product patterns in $\mathbb{C}$. $\mathbb{A}$ (add-to-cart items) serves as ground truth for reward synthesis.

the original query q_0 , enabling in-context reasoning over session dynamics and user intent.

3.3 GRPO-Based Training

We fine-tune a pre-trained LLM π_θ using **Group Relative Policy Optimization (GRPO)**, which extends policy gradient methods [20, 22] by computing relative advantages across a group of candidates, reducing variance and aligning optimization with business outcomes [1, 21].

3.3.1 Two-Stage Training Protocol.

Supervised Fine-Tuning (SFT). We use a larger teacher LLM (DeepSeek-R1-Distill-Qwen-32B [21]), a Qwen-32B base model distilled via DeepSeek-R1’s reasoning pipeline to generate a high-quality synthetic dataset $\mathcal{D}$. For each instance, the teacher receives the full session context (x) and ground-truth add-to-cart items ($\mathbb{A}$) as privileged information, and generates a Chain-of-Thought (CoT) rationale bridging observed behavior with the conversion outcome. A key constraint is that the reformulated query must be indicative of target items *without being overly specific* by focusing on categories and attributes (e.g., “formal wear”) rather than exact product titles (see Figure 2).

The SFT target $y^* = [\text{Rationale}||\text{Reformulation}]$ trains the student via cross-entropy:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, y^*) \sim \mathcal{D}} \log \pi_\theta(y^* | x)$$

Reinforcement Learning with GRPO. The model generates K candidate reformulations $\{q'_i\}_{i=1}^K$ for each input x , each receiving a scalar reward $R(q'_i | x)$. GRPO computes the advantage relative to the group mean:

$$A(q'_i | x) = R(q'_i | x) - \frac{1}{K} \sum_{j=1}^K R(q'_j | x)$$

The GRPO objective is:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}_{x, q'_i \sim \pi_\theta} \left[\log \pi_\theta(q'_i | x) \cdot A(q'_i | x) \right] + \beta D_{\text{KL}}(\pi_\theta || \pi_{\text{ref}})$$

where π_{ref} is the frozen SFT model and $\beta = 0.01$ anchors the policy to prevent drift from the distilled initialization. We use the low-variance KL penalty formulation to stabilize training without overly constraining policy updates. This critic-free formulation provides computational advantages over actor-critic methods while maintaining stability [21].

3.4 Reward Function Design

An effective reformulation should exhibit higher relevance to add-to-cart items ($\mathbb{A}$) than to merely clicked items ($\mathbb{C}$). Given context $x = (q, \mathbb{A}, \mathbb{C})$, the reward is defined as the softmax probability that q' is relevant to $\mathbb{A}$:

$$R(q' | x) = \frac{1}{|\mathbb{A}|} \sum_{j=1}^{|\mathbb{A}|} \frac{e^{\alpha(q', a_j)/\tau}}{Z_j}$$

where $\alpha(\cdot, \cdot)$ is a relevance scorer, τ is a temperature parameter, and Z_j normalizes across all query-item relevance pairs for item a_j :

$$Z_j = \sum_{x \in \{q, q'\}} \sum_{y \in \{\mathbb{A}, \mathbb{C}\}} e^{\alpha(x, y)/\tau}$$

This formulation explicitly trains the model to generate q' maximizing relevance to the strongest intent signals ($\mathbb{A}$).

3.5 Inference

At inference time, the model generates K candidate reformulations $\{q'_i\}_{i=1}^K$ using temperature 0.6 and top- p 0.95 sampling, and selects the candidate with the highest model log-likelihood:

$$q^* = \arg \max_{q'_i} \log \pi_\theta(q'_i | x).$$

Crucially, the reward function R and ground-truth add-to-cart items $\mathbb{A}$ are not used at serving time; they are used exclusively during GRPO training and offline evaluation. All baselines are evaluated identically: each model generates its reformulation, which is then scored by the evaluation relevance model against held-out ground-truth items.

Distinction Between SFT and RL Constraints. The over-specificity guideline (Figure 2) governs the *SFT phase*, preventing the teacher from hardcoding exact product titles or SKU-level identifiers that would not generalize across the retrieval pool. The *RL phase*, by contrast, is designed to exploit statistically reliable behavioral signals including price range, style, and category preferences when these are consistently present across a session. For example, if a user’s add-to-cart items are consistently priced below \$35, the model may infer budget sensitivity as a conversion signal. This is categorically different from arbitrary product-level specificity: it reflects a latent preference grounded in repeated behavioral evidence.

Importantly, reformulated queries are used for *ranking* rather than hard Boolean filtering in production. Items slightly outside inferred constraints are demoted rather than excluded, preserving recall while improving precision for high-intent results.

4 Experiments

4.1 Dataset and Experimental Setup

Proprietary Real-World E-Commerce Dataset. We evaluate our IDLQR on a large-scale proprietary e-commerce dataset comprising 95,000 anonymized customers and approximately 820,000 search sessions across diverse product categories including electronics, fashion, home goods, and beauty products. The dataset captures complete customer journeys including search queries, product clicks, add-to-cart actions, and purchase events, with temporal information enabling reconstruction of sequential user behavior.

To ensure sufficient behavioral signals for meaningful analysis while maintaining data quality, we apply several filtering criteria: (1) sessions must contain at least three distinct search queries to enable longitudinal pattern analysis; (2) each session must include at least one product interaction (click or add-to-cart) to provide behavioral signals; and (3) sessions with obvious bot activity or testing traffic are excluded. The resulting dataset represents authentic user journeys with rich behavioral patterns that reflect realistic shopping intent evolution.

4.2 Training Framework and Implementation

IDLQR follows the two-stage training procedure described in Section 3. The Qwen-0.6B student is first distilled from the DeepSeek-R1-Distill-Qwen-32B teacher and subsequently optimized using GRPO. During GRPO, eight candidate rewrites are generated per input and scored using the conversion-oriented reward described in Section 3. Training is performed on NVIDIA A100 GPUs using mixed-precision (bf16) and gradient checkpointing. The complete training process requires approximately 120 GPU hours, with 80 hours for knowledge distillation and 40 hours for GRPO optimization.

4.3 Baseline Methods and Comparative Framework

We establish a comparative framework to evaluate IDLQR against prompting- and scaling-based alternatives:

- **Zero-shot Qwen-0.6B:** Serves as our minimal baseline, representing the model’s native capability without specialized training
- **Zero-shot Qwen-0.6B with Chain-of-Thought:** Evaluates the impact of explicit reasoning processes without parameter updates
- **Larger zero-shot models (Qwen-1.7B, Qwen-4B, Qwen-8B):** Assesses the performance gains achievable through pure model scaling
- **IDLQR (our full approach):** Combines distillation with GRPO-based optimization

This comparison evaluates the full IDLQR framework against prompting- and scaling-based alternatives, providing insight into the performance-efficiency trade-offs of task-specific query reformulation.

4.4 Evaluation Methodology

We employ three ranking-based metrics at $k \in \{1, 3\}$:

- **MAP@k:** Measures precision of top-k results
- **NDCG@k:** Accounts for relevance and position

- **MRR@k:** Measures how quickly users find relevant items
- All metrics compare original and reformulated queries to quantify improvements.

Relevance Scoring as Conversion Proxy. The evaluation targets are derived from observed historical customer add-to-cart actions rather than generated by an LLM. The evaluation model is used to score the relevance of reformulated queries against these held-out behavioral targets. We use relevance scoring as a proxy for conversion potential, with architectural decoupling to prevent evaluation circularity. The *Reward Scorer* used during GRPO training is a lightweight Transformer model optimized for high-throughput reward estimation. The *Evaluation Model* used to produce the results in Table 1 is a separate, high-capacity production relevance model with distinct architecture and parameter scale. Because the training-time “coach” and evaluation-time “judge” are independent, this architectural separation reduces the risk that reported gains result from overfitting to a single scorer’s biases.

Both models are ultimately anchored in historical user actions (add-to-cart events), providing an exogenous behavioral ground truth.

Evaluation Protocol and Robustness. For each test instance, we generate reformulated queries, execute search using production infrastructure, and compute relevance scores for top impressions. We perform five independent runs with different random seeds and report mean values. Standard deviations are below 0.002 for all reported metrics, indicating stable performance across runs.

5 Results and Discussion

Table 1 presents comprehensive performance metrics for our IDLQR framework compared against all baselines across the evaluation metrics at $k=1$ and $k=3$; consistent improvements were observed across different values of k .

5.1 Overall Effectiveness

IDLQR-0.6B achieves the best performance across all metrics and cutoffs. At $k=1$, MAP improves from 0.2548 to 0.2750 (+7.92%), compared to only +4.40% for the zero-shot Qwen3-0.6B baseline. The gains are consistent across all ranking dimensions: NDCG improves by 7.92% at $k=1$ and 7.50% at $k=3$, while MRR improves by 7.92% and 6.26% respectively, confirming that reformulated queries improve precision, ranking quality, and the speed at which users find relevant products.

At $k=3$, the improvements are even more pronounced, with MAP rising by 8.34% (0.2746→0.2975), NDCG by 7.50% (0.3295→0.3542), and MRR by 6.26% (0.3740→0.3974). This demonstrates that benefits extend throughout the top ranking positions, not just the first result. The largest baseline (Qwen3-8B) achieves comparable MAP@1 (0.2748, +7.84%) but at substantially higher computational cost, as discussed below.

Notably, while all models benefit from reformulation, the magnitude of improvement varies significantly. Larger zero-shot models capture richer semantics, yet targeted distillation and reinforcement learning yield comparable or superior gains with far fewer parameters.

Table 1: Performance comparison of query reformulation approaches across multiple evaluation dimensions. Our IDLQR model achieves superior performance on most metrics despite its compact size. Bold=best; underline=second-best.

k	Model	MAP		NDCG		MRR	
		Original	Enhanced	Original	Enhanced	Original	Enhanced
$k=1$	IDLQR-0.6B (ours)	0.2548	0.2750	0.2548	0.2750	0.2548	0.2750
	Qwen3-0.6B-Nonthinking	0.2548	0.2660	0.2548	0.2661	0.2548	0.2661
	Qwen3-0.6B-Thinking	0.2548	0.2662	0.2548	0.2662	0.2548	0.2662
	Qwen3-1.7B-Nonthinking	0.2548	0.2663	0.2548	0.2663	0.2548	0.2663
	Qwen3-4B-Nonthinking	0.2548	0.2671	0.2548	0.2671	0.2548	0.2671
	Qwen3-8B-Nonthinking	0.2548	<u>0.2748</u>	0.2548	<u>0.2748</u>	0.2548	<u>0.2748</u>
$k=3$	IDLQR-0.6B (ours)	0.2746	0.2975	0.3295	0.3542	0.3740	0.3974
	Qwen3-0.6B-Nonthinking	0.2746	0.2831	0.3295	0.3384	0.3740	0.3849
	Qwen3-0.6B-Thinking	0.2746	0.2832	0.3295	0.3385	0.3740	0.3851
	Qwen3-1.7B-Nonthinking	0.2746	0.2832	0.3295	0.3385	0.3740	0.3851
	Qwen3-4B-Nonthinking	0.2746	0.2841	0.3295	0.3395	0.3740	0.3863
	Qwen3-8B-Nonthinking	0.2746	<u>0.2967</u>	0.3295	<u>0.3530</u>	0.3740	<u>0.3967</u>

5.2 Scaling vs. Task-Specific Optimization

A striking finding is the non-linear relationship between model size and reformulation quality. Scaling from 0.6B to 4B without specialized training yields only marginal MAP@1 gains (0.2660→0.2671, +0.43% absolute), with MAP@1 progressively improving from 0.2660 (0.6B) to 0.2663 (1.7B) to 0.2671 (4B) to 0.2748 (8B). While larger models do capture more nuanced query understanding, our 0.6B IDLQR (0.2750) achieves comparable performance with the 8B model (0.2748) despite being 13.3× smaller (~1.2GB vs. 16GB), providing a favorable performance-efficiency trade-off at a fraction of the model size. This pattern holds at $k=3$, where IDLQR achieves 0.2975 MAP compared to 0.2967 for Qwen3-8B.

These results suggest that knowledge distillation followed by GRPO optimization provides a deployment-efficient alternative to raw parameter scaling for this setting. Distillation provides the compact student with a task-specific initialization, while GRPO subsequently refines the model under the conversion-oriented reward objective.

5.3 Business Implications

From a practical perspective, improvements at top-ranked positions are particularly relevant for e-commerce search, where customer interaction is concentrated near the top of the ranking. Importantly, IDLQR achieves these offline ranking improvements with a compact model and sub-100ms inference latency, making it suitable for latency-constrained deployment. Whether these improvements translate into gains in conversion or revenue requires validation through online experimentation.

Overall, the results suggest that conversion-oriented task-specific optimization can provide a competitive alternative to increasing model scale for query reformulation. Crucially, IDLQR achieves this with a compact 0.6B model maintaining sub-100ms latency, making it suitable for latency-constrained production environments.

Diversity and Long-Term Ecosystem Health. Unlike embedding-based retrieval that can collapse into narrow semantic clusters, IDLQR performs attribute-level reasoning (e.g., inferring “breathable fabric” from “gym gear” clicks), naturally expanding the search space to relevant but lexically diverse products. Additionally, IDLQR-generated queries feed into a downstream ranker that incorporates standard diversity and exploration filters, ensuring long-term catalog coverage and ecosystem health are maintained alongside conversion optimization.

6 Conclusion

This work introduced IDLQR, a two-stage framework that combines large-teacher knowledge distillation with reinforcement learning using Group Relative Policy Optimization (GRPO) to optimize customer search journeys in e-commerce platforms. By explicitly prioritizing conversion-driven signals, particularly add-to-cart actions over simple clicks, IDLQR bridges the gap between user intent understanding and purchase-oriented ranking.

Extensive experiments on large-scale real-world e-commerce data demonstrate that a lightweight Qwen-0.6B model, when distilled from a DeepSeek-R1-Distill-Qwen-32B teacher and further fine-tuned with GRPO, achieves performance comparable to zero-shot models up to 8B parameters in conversion-focused metrics while maintaining sub-100 ms latency and a small memory footprint suitable for production deployment. These results suggest that task-aligned optimization and reward-driven fine-tuning can provide a competitive alternative to model scaling for latency-constrained query reformulation.

While our offline evaluation uses an architecturally independent relevance model as a proxy for downstream conversion, offline evaluation cannot fully substitute for online business validation. Future work will evaluate IDLQR through online A/B experiments measuring conversion, engagement, and revenue impact in production. We

also plan to expand the reward design to incorporate long-term user engagement, multi-session personalization, and dynamic real-time feedback signals, as well as extend this approach to other verticals beyond search, including personalized recommendations and promotions.

7 Reproducibility Details

7.1 GRPO Hyperparameters

Table 2: Complete hyperparameter configuration for GRPO training.

Parameter	Value
Candidate rewrites per input (K)	8
Training temperature (τ)	1.0
Inference temperature	0.6
Inference top- p	0.95
KL penalty coefficient (β)	0.01
KL penalty type	low_var_kl
Learning rate (η)	1e-6
Optimizer	AdamW
Weight decay	1e-2
Max gradient norm	1.0
Precision	bf16
Training epochs	15
Validation frequency	Every 5 epochs
Tensor parallelism	8
GPU memory utilization	60%
Hardware	8× NVIDIA A100 (1 node)
Total training time	~120 GPU-hours
Knowledge distillation	~80 GPU-hours
RL optimization	~40 GPU-hours

7.2 Data Splits and Statistics

The dataset is split 70/15/15 into training (574K sessions), validation (123K sessions), and test (123K sessions) partitions. Sessions are split at the customer level to prevent data leakage.

7.3 Cold-Start Handling

For users with fewer than 3 interactions in the current session, the system defaults to the original query q_0 without reformulation, ensuring stable behavior for cold-start scenarios.

7.4 Deployment Details

IDLQR operates as a pre-retrieval query rewriting service. The system serializes the user session context, generates q^* via the Qwen-0.6B student model, and passes q^* to both lexical (BM25) and semantic retrieval stages. Inference latency is maintained below 100ms on a single GPU. For sessions below the interaction threshold, the pipeline falls back to q_0 transparently.

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